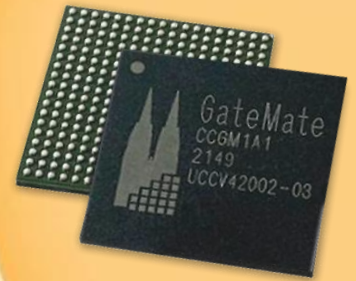


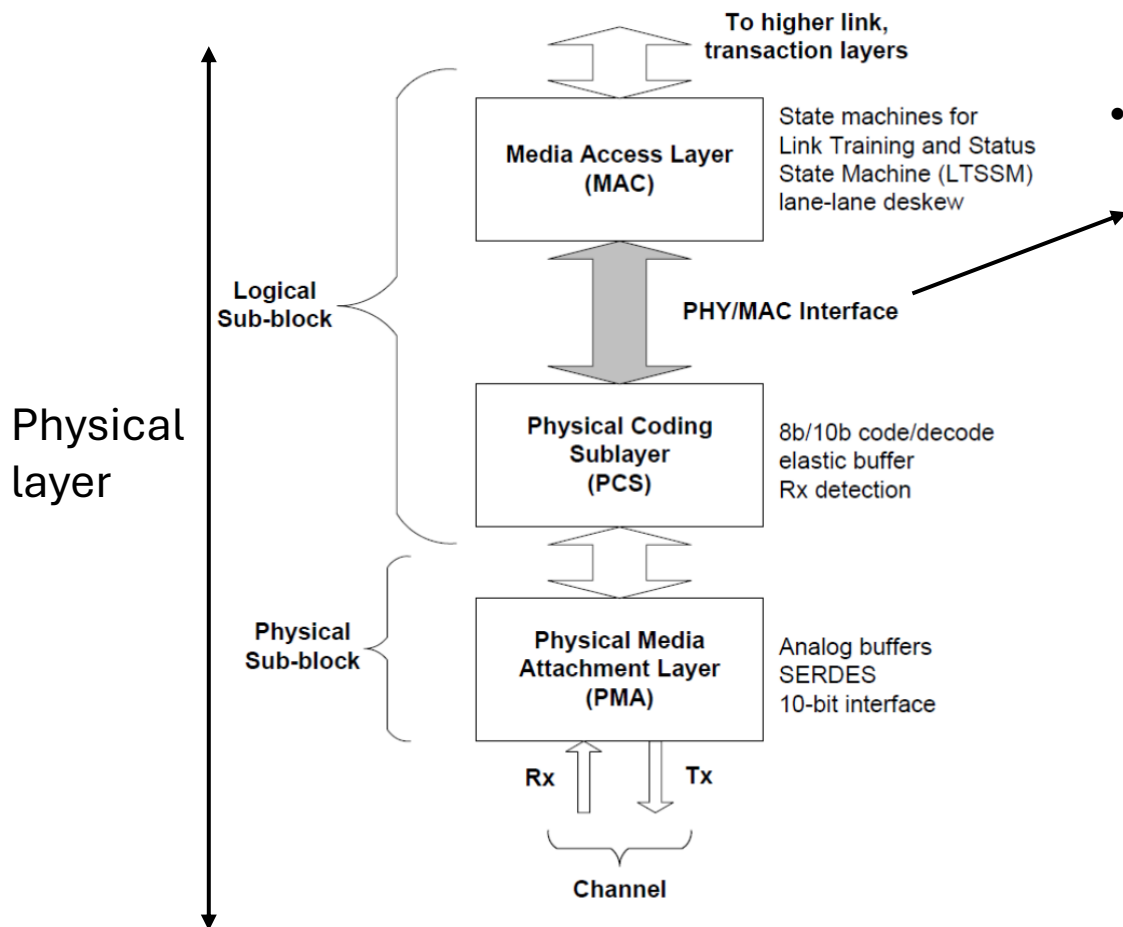
GateMate™ FPGA

MAC layer

Vinh Hai Luong, B.Sc.
Research&Development
Cologne Chip AG

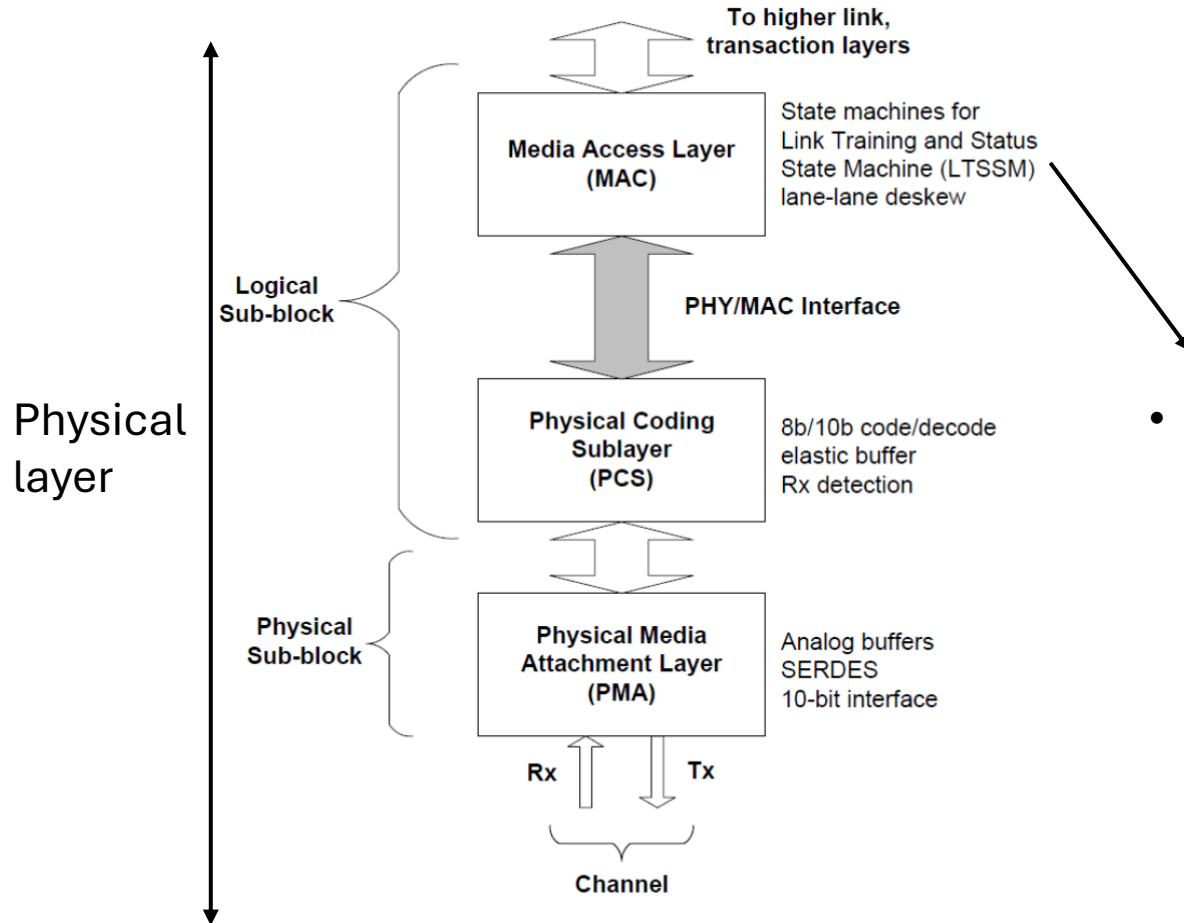


PIPE Overview



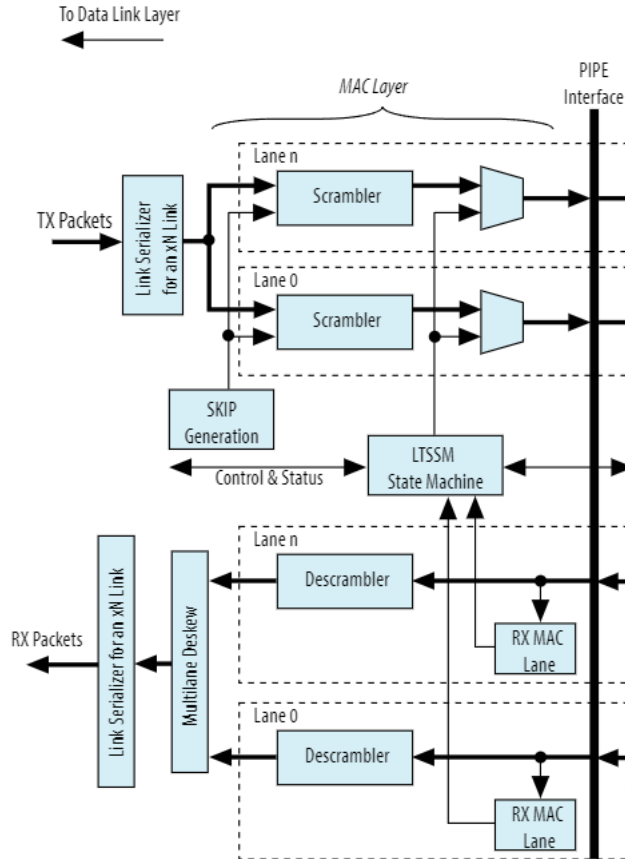
- The **PHY Interface for the PCI Express Architecture (PIPE)** is part of the physical layer and serves as an interface between the Media Access Layer (MAC) and the Physical Coding Sublayer (PCS).

MAC Overview

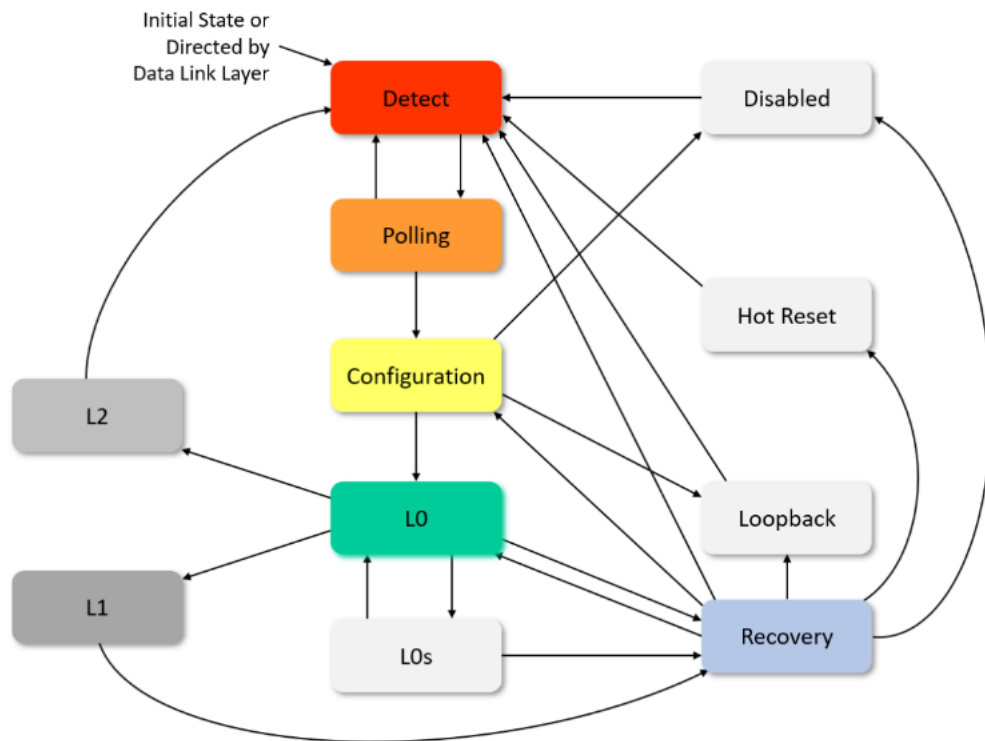


- The **Media Access Layer (MAC)** is part of the physical layer. It lies in between the **PIPE Interface** and the **Data Link Layer (DLL)**.

MAC Overview

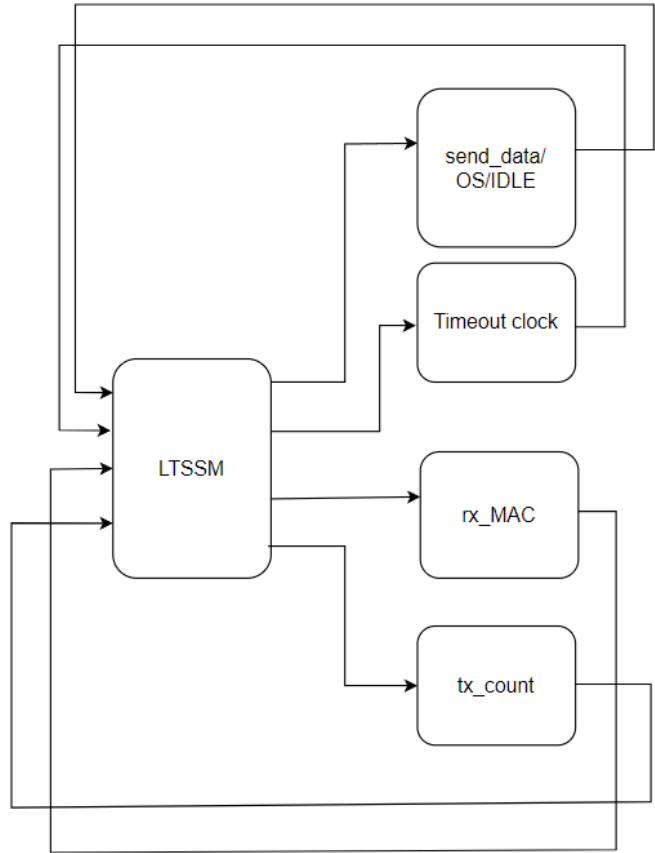


- Scrambler/Descrambler to decode/encode data.
- SKP Generator to insert SKP Ordered Set at an interval between 1180 and 1538 Symbol Times, compensating different bit rates of two ports.
- LTSSM to control Link training and get the Link up.



LTSSM Block Diagram

LTSSM Logic



- Send_data/OS/IDLE: Assemble the transmitted data, e.g. TS1 and TS2 Ordered Sets.
- Timeout clock: Trigger timeout clock at each state according to the Specs.
- Rx_MAC:
 - Detect Link/Lane number from received Ordered Sets.
 - Check if received Ordered Sets are valid based on previously determined Link/Lane number.
 - Count the number of consecutive correctly received Ordered Sets and feedback the count flag to the LTSSM.
 - Detect and remove SKP Ordered Sets from received sequence.
- Tx_count: Count the number of transmitted Ordered Set and feedback the count flag to LTSSM

References

- [PHY Interface for the PCI Express* Architecture \(PIPE\) Specs](#)
- [Simon Southwell's PCIe Primer](#)
- [PCI Express* Solutions User Guide](#)

Contact

Luong Vinh Hai, B.Sc.

Cologne Chip AG

Eintrachtstr. 113

50668 Köln

